Math 199, Spring 2026
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Week 4 - Sets, relations, and functions

1) Let S^1 be the (set of points on the) standard unit circle in the plane.

(a) Sketch a picture of S^1.

(b)Describe the set S^1 as a relation on $\mathbb{R}$.

(c) Is this relation a function? Prove why or why not.

(d) Describe one or two ways to modify S^1 to turn it into a function. Explain this both in terms of your experience with functions from Calculus and in terms of the language from Math 314.

2) **Definition:** Let $S \subseteq \mathbb{R}$. A function $f : S \to \mathbb{R}$ is *bounded* if $\exists M \in \mathbb{R}$ such that $|f(x)| \leq M$ for all $x \in S$.

(a) Define what it means for a function to be *unbounded* (i.e. negate the definition).

(b) Give an example of a function which is bounded and one which is unbounded (no proof required).

(c) Consider the statement obtained by switching the quantifiers: For all $x \in S$, $\exists M \in \mathbb{R}$ such that $|f(x)| \leq M$. Prove that this statement is true for every function f.

(d) Is the function $\arctan : \mathbb{R} \to \mathbb{R}$ bounded or unbounded? Prove your answer.

(e) Is the function $\mathrm{Sq} : \mathbb{R} \to \mathbb{R}$, defined by $\mathrm{Sq}(x) = -x^2$, bounded or unbounded? Prove your answer.

(f) Is the function $\mathrm{Sq} : (-2026, 2026) \to \mathbb{R}$, defined as in (e), bounded or unbounded? Prove your answer.

3) Let $f : \mathbb{R} \to \mathbb{R}$ be a function. You can suppose that the value of f at $x \in \mathbb{R}$ can be described by some familiar formula, $f(x)$.

(a) Is it true that the formula $f(x)$ always describes a function $f^{\mathbb{Q}} : \mathbb{Q} \to \mathbb{R}$? Prove why or why not.

(b) Is it true that the formula $f(x)$ always describes a function $f^{\mathbb{Q}}_{\mathbb{Q}} : \mathbb{Q} \to \mathbb{Q}$? Prove why or why not.

(c) Is it true that the formula $f(x)$ always describes a function $f_{\mathbb{Q}} : \mathbb{R} \to \mathbb{Q}$? Prove why or why not.

(d) Does there exist a function $f : \mathbb{R} \to \mathbb{R}$ for which $f^{\mathbb{Q}}$ and $f_{\mathbb{Q}}^{\mathbb{Q}}$ are functions, but $f_{\mathbb{Q}}$ is not? Give an example of such an f, or prove that no such function exists.

(e) Does there exist a function $f : \mathbb{R} \to \mathbb{R}$ for which $f^{\mathbb{Q}}$ and $f_{\mathbb{Q}}$ are functions, but $f_{\mathbb{Q}}^{\mathbb{Q}}$ is not? Give an example of such an f, or prove that no such function exists.

(f) Does there exist a function $f : \mathbb{R} \to \mathbb{R}$ for which $f^{\mathbb{Q}}$, $f_{\mathbb{Q}}^{\mathbb{Q}}$, and $f_{\mathbb{Q}}$ are all functions? Give an example of such an f, or prove that no such function exists.

4) Let $f : \mathbb{R} \to \mathbb{R}$ be a function, and let $f^{\mathbb{Q}}$, $f_{\mathbb{Q}}^{\mathbb{Q}}$, and $f_{\mathbb{Q}}$ be as in problem (3). For each of the examples below, determine whether each of $f^{\mathbb{Q}}$, $f_{\mathbb{Q}}^{\mathbb{Q}}$, and $f_{\mathbb{Q}}$ is a function or not, and justify your answer.

(a) $f(x) = x^2$

(b) $f(x) = \sin(\pi x)$

(c) $f(x) = 17$

(d) $f(x) = \sqrt{x}$

(e) $f(x) = \sqrt{2}$

(f) $f(x) = \begin{cases} \dfrac{x^3 - 4x^2 + 1}{2026}, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$

5) **Definition:** A relation $\square$ on a set S is *transitive* if $\forall a, b, c \in S$, we have $(a \square b \ \wedge \ b \square c) \implies (a \square c)$.

(a) Give an example of a relation that is transitive, and give an example of a relation that is not transitive.

(b) Give an example of a relation that is both a (non-identity) function and transitive.